Evidence-grounded belief revision as a prerequisite for entropy-production measurements in decentralized LLM-agent networks

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Abstract

Statistical-mechanical descriptions of language-model agents require more than assigning binary spins to generated choices. In particular, an empirical transition kernel is meaningful only if communicated evidence measurably and reproducibly changes an agent's local belief distribution. We introduce a controlled binary latent-state environment in which private and communicated signals have explicit likelihoods, responses separate continuous belief from action and commitment, and message effects are estimated with matched prompts and cluster-level uncertainty. A prospectively frozen experiment with Qwen2.5-7B-Instruct comprised 864 decisions in 48 independent matched information clusters. Delivered evidence caused a positive average signed change in reported belief log odds, 0.1343 (95% cluster-bootstrap interval 0.0915–0.1811), with positive application-specific intervals in two semantic framings. Nevertheless, the response was nonmonotone in source reliability, binary choices were strongly right-skewed, and reported probabilities were poorly calibrated to repeated choices. These failures triggered a prospective stop before reciprocal or nonreciprocal network trajectories were generated. The result therefore does not estimate entropy production in an LLM network. Instead, it identifies continuous evidence response, directional symmetry, calibration, and transition diversity as necessary qualification conditions for defensible coarse-grained stochastic thermodynamics of decentralized LLM agents.

1 Introduction

Heat-bath dynamics, strategic logit response, and stochastic thermodynamics provide a natural mathematical language for interacting decision makers [1–3]. Directed interactions can generate stationary probability currents and positive entropy production even when the reciprocal limit admits a Gibbs distribution [4–6]. Large language models (LLMs), however, introduce private context, hidden memory, linguistic framing, and an elicitation layer between the latent policy and the recorded state. A binary answer that does not switch after a message is consequently ambiguous: the message may have been ignored, may have caused a subthreshold probability update, or may have been overwhelmed by an option bias.

Recent work has begun to quantify collective alignment and conformity in LLM populations [7–9], while observable belief revision is itself an active measurement problem

[10]. These studies do not remove the need to identify the local response kernel before applying pathwise irreversibility estimators. Under coarse graining, hidden degrees of freedom can make an observed entropy-production estimate only a lower bound or can invalidate a first-order Markov interpretation [11–13].

This work reports a prospective qualification study. Its purpose was to open a decentralized reciprocal/nonreciprocal LLM-network experiment only if an LLM used likelihood-bearing messages with sufficient monotonicity and transition diversity. The qualification found an average continuous belief effect but failed two frozen requirements. We preserve that no-go outcome and use it to specify the measurement boundary, rather than estimating a network quantity from an inadequate empirical state.

2 Statistical-mechanical reference and measurement boundary

The immutable preceding study defined a coupled belief–action microstate $x = (\mathbf{b}, \mathbf{a})$ with $b_i, a_i \in \{-1, +1\}$ and reciprocal energy

$$H(x) = -\frac{J_b}{2}\sum_{ij} A_{ij}^{(c)} b_i b_j - \frac{J_a}{2}\sum_{ij} A_{ij}^{(d)} a_i a_j - K\sum_i a_i b_i - \sum_i h_i b_i - \sum_i g_i a_i. \tag{1}$$

With symmetric couplings, a static environment, common decision temperature, and asynchronous heat-bath updates, the finite-state process obeys detailed balance with $\pi(x) \propto \exp[-H(x)/T]$. Directed communication was parameterized as a symmetric component plus controlled antisymmetric perturbation. For a transition kernel $W_\alpha = W_0 + \alpha V + O(\alpha^2)$ around the reversible reference, the stationary probability current is first order in α, while its affinity is also first order. Their product gives the quadratic leading entropy production

$$\sigma(\alpha) = \frac{1}{2}\sum_{x,y} J_{xy}(\alpha) \log \frac{\pi_\alpha(x) W_\alpha(x,y)}{\pi_\alpha(y) W_\alpha(y,x)} = C\alpha^2 + O(\alpha^3). \tag{2}$$

That identity is exact for the specified finite Markov reference up to the stated perturbative order; it is not automatically an identity for LLM text generation.

For LLM agents, the recorded state was intended to be $s_i = (p_i, b_i, a_i, c_i, m_i)$: reported probability, derived belief, typed action, commitment, and bounded memory. A network entropy-production claim would additionally require stationarity or a declared driven protocol, adequate state closure, nondegenerate transition estimates, and bias controls. V11 tests the prerequisite local evidence response before attempting those conditions.

3 Independent LLM-agent architecture

Each agent has a persistent identity, private signal and probability, independent bounded memory, typed action authority, commitment, inbox, and outbox. Delivered packets are the only route by which peer evidence enters a context. The scheduler chooses an update opportunity and validates a typed tool; it does not select the reported probability, belief, action, commitment, or send/abstain decision. Evaluator-only latent state is excluded from prompts. Counterfactual tests modify one private vault while checking byte-identical peer state.

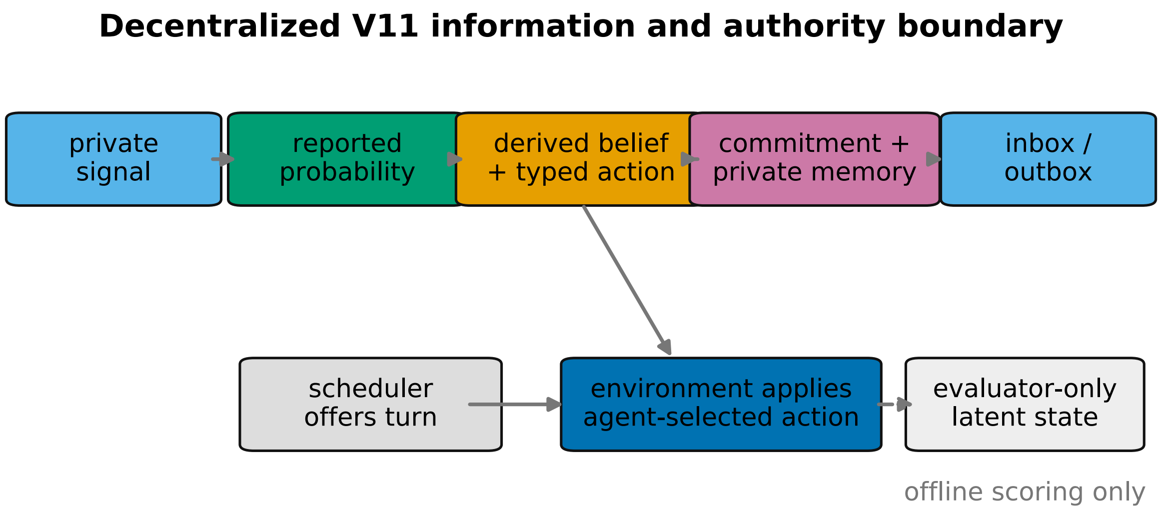

Figure 1: V11 agent information and authority boundary. The evaluator state is available only for offline scoring.

The primary model is `Qwen/Qwen2.5-7B-Instruct` [14]. Its pinned revision is `a09a35 458c702b33eeacc393d103063234e8bc28`; it is loaded with NF4 double quantization and BF16 computation. Sampling temperature is 0.65, top-p is 0.90, and at most one bounded JSON repair is allowed. The inference sampling temperature is not interpreted as the effective decision temperature in Eq. (1).

4 Evidence-generating process

Let $\theta \in \{L, R\}$ be a hidden evaluator state. Agent i observes a conditionally independent signal s_i with declared reliability r_i,

$$P(s_i = \theta \mid \theta, r_i) = r_i, \qquad r_i \in \{0.55, 0.65, 0.75, 0.85\}. \tag{3}$$

The normative log-likelihood increment for an observation favouring R is $\log[r/(1-r)]$ and changes sign for an observation favouring L. It is a reference, not an instruction to emit a Bayesian posterior. Packets include source identity, observation, reliability, observation and delivery times, freshness, domain, and deterministic wire serialization.

Two application-neutral semantic frames instantiate the same probability model: route viability and repair hypothesis. Belief is elicited before action and commitment. The structured response contains `probability_right`$\in [0, 1]$; b_i is then derived using a frozen 0.5 threshold. Repeated independent generations from identical information states provide an empirical binary-choice frequency separate from the reported probability.

5 Prospective protocol and estimands

An engineering pilot selected only interface and practical gate values. The decisive protocol was then frozen at SHA-256 `7bd4082f9d085222e22d195e3ff603f0f76e27c36b347bfc41 32fbb164a3d03f`. It specified 48 matched information clusters, three prompt paraphrases,

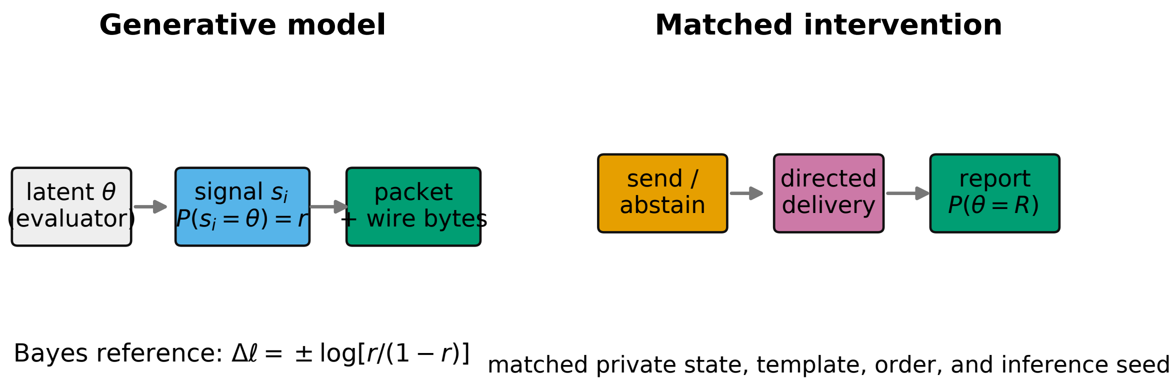

Figure 2: Explicit signal model and matched evidence intervention. Private state, prompt template, option order, and inference seed are held fixed.

counterbalanced option order, two samples per condition, four reliabilities, and 864 requests. The independent unit is the information cluster, not the generation or token.

For response p_{ic} in condition c and its no-message matched response p_{i0}, the primary row effect is

$$d_{ic} = q_c\{\operatorname{logit}(p_{ic}) - \operatorname{logit}(p_{i0})\}, \qquad q_c \in \{-1, +1\}, \tag{4}$$

oriented toward the evidence. Cluster means are bootstrapped with 10,000 cluster resamples. A direction-balanced placebo contrast removes a common message-presence shift while retaining its magnitude as a diagnostic.

Progression required all components: first-pass and repaired validity, positive directional and placebo-adjusted intervals with estimates at least 0.10, positive intervals in both frames, LLR-response slope at least 0.05 with a monotonic tolerance of 0.03, bounded order/paraphrase effects, continuous standard deviation at least 0.05, each binary choice in at least 10% of valid responses, prompt isolation, and accepted agent-selected messages. Failure locked the formal network stage.

6 Qualification results

There were 842 first-pass-valid responses, 21 repaired-valid responses, and one invalid response after repair. First-pass and final validity were 97.45% and 99.88%. The primary signed effect was 0.1343 (95% cluster-bootstrap interval 0.0915–0.1811), and the placebo-adjusted estimate was 0.1343 (0.0905–0.1816). Route viability yielded 0.1886 (0.1164–0.2684); repair hypothesis yielded 0.0799 (0.0445–0.1197).

The reliability-specific means were 0.0463, 0.1686, 0.1779, and 0.1388 at $r = 0.55$, 0.65, 0.75, and 0.85. The fitted slope against normative LLR was 0.04965, below the frozen 0.05 minimum; the decline at the highest reliability exceeded tolerance. The right-choice fraction was 0.9224, above the frozen 0.90 ceiling. Thus reliability monotonicity and transition diversity failed, and the formal experiment did not run.

A post-qualification diagnostic, excluded from the progression decision, localized substantial directional asymmetry. Right-supporting packets had a mean signed change of

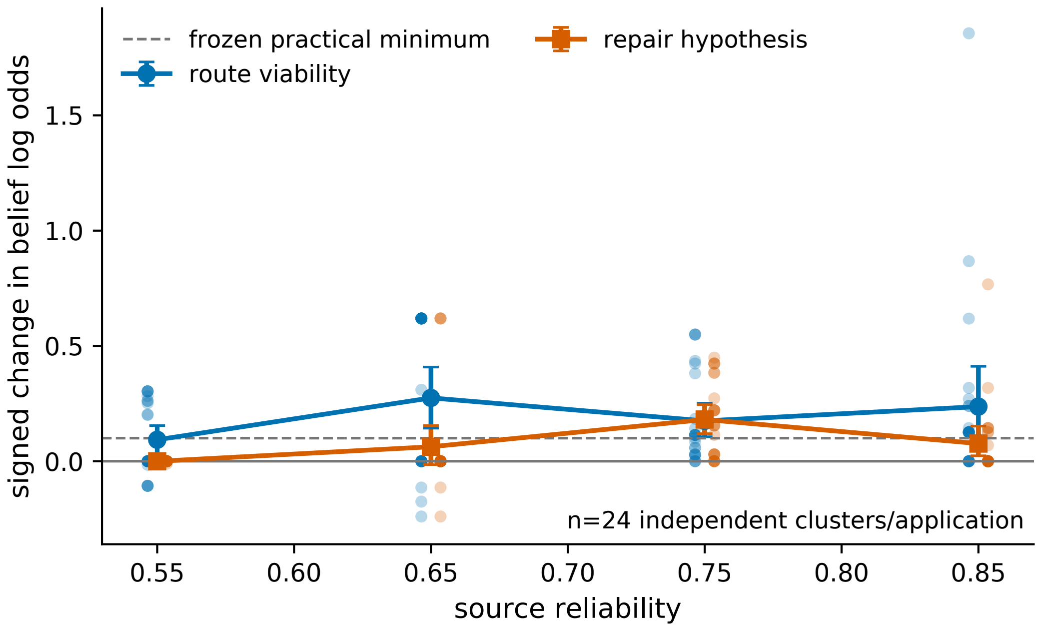

Figure 3: Cluster-level signed log-odds response by source reliability. Points are independent cluster means; symbols show means and 95% cluster bootstrap intervals. The dashed line is the frozen practical minimum, not a fitted threshold.

0.5574, while left-supporting packets had −0.2939. The pooled positive effect therefore does not establish symmetric evidence integration.

6.1 Reported probability versus empirical choice

Across 432 repeated-information cells, each with two independent samples, mean reported $P(R)$ was 0.6697 whereas the empirical right-choice frequency was 0.9225. The Brier score was 0.0968 and expected calibration error was 0.2955. Two repetitions per cell are too few for a high-resolution calibration curve, but the global discrepancy and choice degeneracy are sufficient to reject use of the reports as calibrated transition probabilities.

7 Why entropy production was not estimated

Positive average message response is necessary but not sufficient for a network estimator. The frozen formal design would have compared reciprocal and degree/traffic-matched directed networks, tested increasing history depth, and estimated block time-reversal divergence with synthetic-chain bias controls. The qualification instead revealed (i) nonmonotone reliability response, (ii) a nearly saturated right-choice process, (iii) poor calibration, and (iv) post-gate direction asymmetry. Estimating currents from that binary state would conflate option bias with evidence flow and produce unstable zero-count corrections. Calling such a quantity exact entropy production would be particularly misleading because hidden textual memory can violate first-order Markov closure.

Consequently, V11 generated no reciprocal or nonreciprocal LLM trajectories, no empirical stationary distribution, no transition-current entropy production, and no Markov-adequacy estimate. This is a prospective design decision rather than absence of reporting.

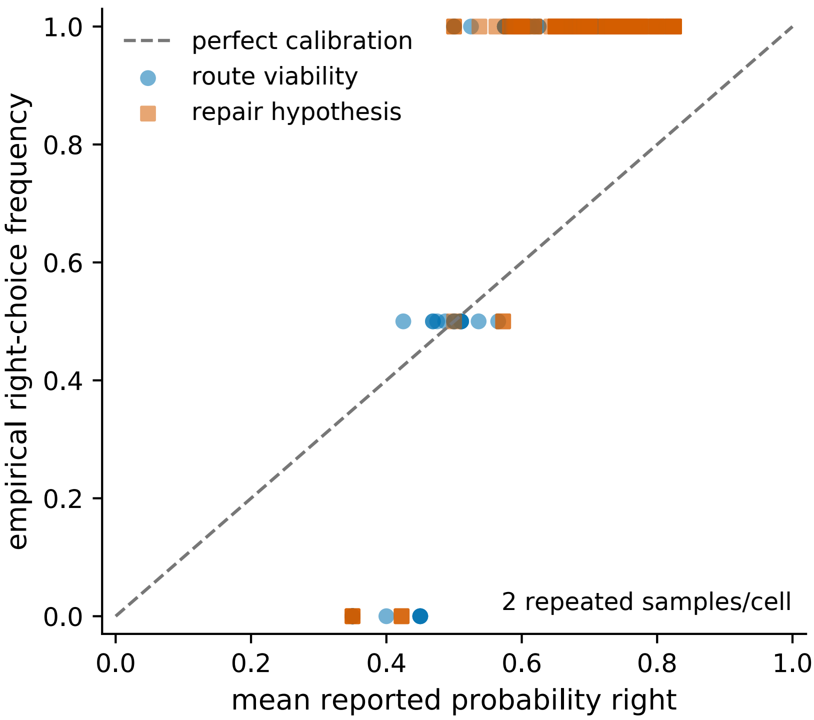

Figure 4: Reported probability and empirical repeated-choice frequency. The figure is descriptive because each information cell has two repetitions.

8 Modular-network anomaly inherited from V10

An audit also resolved a V10 analytical-model anomaly without modifying its frozen files. The modular graph generator held within/between-community edge probabilities fixed as N grew, so mean degree increased while interaction sums were not degree-normalized. Local fields saturated, acceptance collapsed, and trajectories froze. Longer runs did not restore mixing; dividing the adjacency by mean degree restored order-one acceptance and nonzero pathwise estimates. The effect was therefore a topology-construction and metastability artifact, not universal modular scaling. Any future formal LLM study must fix mean degree or normalize coupling prospectively.

9 Discussion

The continuous endpoint repairs the central ambiguity of a binary switch test: subthreshold message effects are now visible. Yet a scalar probability emitted in JSON is not automatically an empirical policy probability. The V11 result shows both sides. Likelihood-bearing messages measurably changed the continuous coordinate in two framings, but the response was biased and could not support a diverse transition kernel.

The boundary has direct implications for statistical mechanics of LLM agents. First, a recorded microstate must be operationally sufficient, not merely easy to parse. Second, reciprocal nulls require local response symmetry and traffic-matched messaging in addition to symmetric graph support. Third, calibration should be estimated from repeated choice frequencies or constrained token probabilities rather than assumed from narrated confidence. Finally, if finite memory materially predicts transitions after conditioning on the nominal state, time-reversal divergence should be described as a coarse-grained lower bound rather than full entropy production.

10 Limitations

The study uses one 7B instruction model, two simulated semantic framings, two repetitions per information cell, and self-reported probabilities. Paired seeds reduce but cannot eliminate stochastic generation effects. There are no dynamic networks, no second model, no application performance outcomes, and no human participants. The post-gate direction analysis is exploratory. The inherited analytical heat-bath theory is exact for its finite model but does not validate an LLM transition law.

11 Conclusion

Calibrated delivered evidence produced a detectable average continuous belief shift, resolving the identifiability problem of the preceding binary test. However, monotonicity and transition diversity failed prospectively, and reported probabilities were not calibrated to empirical choices. V11 therefore stops before measuring entropy production in an LLM-agent network. Its positive contribution is a falsifiable qualification framework and a clear measurement boundary for future statistical-mechanical studies of decentralized LLM agents.

Data and software availability

Source, frozen configuration, tests, aggregate tables, and vector figures are in the V11 repository-facing package. Raw prompts and completions are excluded from Git and preserved under `/workspace/ThermoAgent-v11-artifacts/`; directory tree checksums are in `results/llm_agent_entropy_v11/reproducibility/`. The exact model revision, environment versions, seeds, protocol hash, and reproduction commands are recorded in the accompanying README. No model weights are redistributed.

A Frozen gate disposition

Component	Result	Main diagnostic
Structured validity	pass	97.45% first pass; 99.88% repaired
Directional effect	pass	0.1343; CI excludes zero
Placebo-adjusted effect	pass	0.1343; CI excludes zero
Semantic replication	pass	positive interval in both frames
Reliability monotonicity	fail	slope 0.04965; high-r decline
Transition diversity	fail	92.24% right choices
Order/paraphrase sensitivity	pass	within frozen bounds
Authority and prompt isolation	pass	deterministic tests and fingerprints

Table 1: Every component was required for progression.

B Computational accounting

The decisive qualification used 886 model calls, 559,895 prompt tokens, 76,620 generated tokens, and 1,536.064 seconds of measured generation latency on an RTX 4090. The retained pilot plus qualification used 1,017 calls and 0.4882 measured generation GPU-hours. Three invalidated engineering pilots are preserved externally; the earliest provider did not retain token usage for responses invalid after repair, so all-pilot token totals are reported only as documented lower bounds.

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